

error $\text{TV}(P_{\text{data}}, P_{t_0})$ increases due to the heavier injected Gaussian noise. Under a training horizon of $T = \tilde{\mathcal{O}}(\log n)$, the choice of $t_0 = n^{-\frac{1-\delta(n)}{k+5}}$ optimally balances the early stopping error and the score estimation error.

Eigenspace Estimation using Generated Samples. The latent subspace estimation in Theorem 3 shows that the subspace can be accurately recovered with high probability. Specifically, generating $\tilde{\mathcal{O}}(dn^{\frac{2(1-\delta(n))}{k+5}} \log n)$ samples from the trained diffusion model ensures that the eigenvalues and eigenspace of the sample covariance matrix $\hat{\Sigma}_0$ closely approximate those of Σ_0 , with the error proportional to the score estimation error. Moreover, if $\text{Eigen-gap}(k)$ increases—indicating an improvement in the factor model identification, then the estimation error of the k -dimensional eigenspace decreases.

Further Discussion on Dimension Dependence. Our sample complexity bounds in Theorem 3 circumvent the curse of ambient dimensionality d under very mild assumptions, namely, score function being Lipschitz and the distribution of factors being sub-Gaussian. As a result, these bounds characterize learning efficiency being adaptive to the subspace dimension k even in the most challenging scenarios. In practical applications, however, data distributions often possess more favorable regularity properties—such as higher-order smoothness in the score function or the distribution of returns—which may lead to better learning efficiency compared to the theoretical bound. While refining our bounds under such additional properties is beyond the scope of this paper, we present comprehensive numerical results in Sections 6 and 7 to illustrate the strong empirical performance of diffusion factor models, particularly in the “small data” regime.

Proof Sketch. The proof is deferred to Appendix C.1; here, we highlight its main ideas. The outline has two parts: (I) the key steps in establishing the distribution estimation result in Theorem 3, and (II) the technical components for proving the latent subspace recovery results, emphasizing novel coupling and concentration arguments.

(I) Estimation of return distribution. We bound each term in the decomposition (25) separately.

1. Term $\text{TV}(P_{\text{data}}, P_{t_0})$ is the early-stopping error. By direct calculations using the Gaussian transition kernel, we show that it is bounded by $\mathcal{O}(dt_0)$.
2. Term $\text{TV}(P_{t_0}, \tilde{P}_{t_0})$ captures the statistical estimation error. We apply Girsanov’s Theorem (Karatzas and Shreve 1991, Theorem 5.1; Revuz and Yor 2013, Theorem 1.4) to show that the KL divergence $\text{KL}(P_{t_0}, \tilde{P}_{t_0})$ is bounded by the L^2 score estimation error developed in Theorem 2. Further, by Pinsker’s inequality (Tsybakov 2009, Lemma 2.5), we convert the KL divergence bound into a total variation distance bound.
3. Term $\text{TV}(\tilde{P}_{t_0}, \hat{P}_{t_0})$ reflects the mixing error of the forward process (1). Using the data processing inequality (Thomas and Joy 2006, Theorem 2.8.1), we show that it is a non-leading error term of order $\tilde{\mathcal{O}}(\exp(-T))$.

(II) Latent subspace recovery. The crux is to bound the covariance estimation error $\|\hat{\Sigma}_0 - \Sigma_0\|_{\text{op}}$ by the following lemma.

Lemma 2. Assume the same assumptions as in Theorem 3 and take $\hat{\Sigma}_0$ as the estimator in (24) with m samples from Algorithm 1. It holds that, with probability at least $1 - \delta$,

$$\|\hat{\Sigma}_0 - \Sigma_0\|_{\text{op}} = \mathcal{O}\left(\lambda_{\max}(\Sigma_0)(1 + \sigma_{\max}^k) d^{\frac{5}{4}} k^{\frac{k+10}{4}} n^{-\frac{1-\delta(n)}{k+5}} \log^{\frac{5}{2}} n\right). \quad (26)$$

Here, m satisfies

$$m = \mathcal{O}\left(\lambda_{\max}^{-2}(\Sigma_0) d n^{\frac{2(1-\delta(n))}{k+5}} \log n\right). \quad (27)$$

The complete proof of Lemma 2 is deferred to Appendix C.2.1. Using Lemma 2 in combination with Weyl's theorem and Davis-Kahan theorem (Davis and Kahan 1970), we derive the desired results for latent subspace recovery.

Proving Lemma 2 is similar to that for the estimation of return distribution. We upper bound $\|\hat{\Sigma}_0 - \Sigma_0\|_{\text{op}}$ by

$$\|\hat{\Sigma}_0 - \Sigma_0\|_{\text{op}} \leq \underbrace{\|\Sigma_0 - \Sigma_{t_0}\|_{\text{op}}}_{(A)} + \underbrace{\|\Sigma_{t_0} - \tilde{\Sigma}_{t_0}\|_{\text{op}}}_{(B)} + \underbrace{\|\tilde{\Sigma}_{t_0} - \check{\Sigma}_{t_0}\|_{\text{op}}}_{(C)} + \underbrace{\|\hat{\Sigma}_0 - \check{\Sigma}_{t_0}\|_{\text{op}}}_{(D)},$$

where Σ_{t_0} , $\tilde{\Sigma}_{t_0}$, $\check{\Sigma}_{t_0}$ are the covariance of P_{t_0} , $\tilde{P}_{t_0}$ and $\hat{P}_{t_0}$, respectively. Analogous to the upper bound of the total variation distance in (25), term (A) corresponds to the early-stopping error; term (B) captures the statistical estimation error; and term (C) reflects the mixing error. The additional term (D) represents a finite-sample concentration error arising from the use of m samples in Algorithm 1.

We bound each term separately. Term (A) can be bounded by direct calculations using the Gaussian transition kernel; term (D) is bounded using matrix concentration inequalities (Vershynin 2018, Theorems 3.1.1 and 4.6.1). However, bounding terms (B) and (C) requires a novel analysis, as small total variation distances $\text{TV}(P_{t_0}, \tilde{P}_{t_0})$ and $\text{TV}(\tilde{P}_{t_0}, \hat{P}_{t_0})$ do not immediately imply small error bounds on the covariance matrix. In fact, we show the following L^2 bound based on a coupling between two backward SDEs, which converts to bounds on (B) and (C).

Lemma 3. Assume the same assumptions as in Theorem 3. Consider the following coupled SDEs:

$$\begin{cases} d\mathbf{R}_t^{\leftarrow} &= \left(\frac{1}{2}\mathbf{R}_t^{\leftarrow} + \nabla \log p_{T-t}(\mathbf{R}_t^{\leftarrow})\right)dt + d\bar{\mathbf{W}}_t, \quad \text{with } \mathbf{R}_0^{\leftarrow} \sim P_T, \\ d\hat{\mathbf{R}}_t^{\leftarrow} &= \left(\frac{1}{2}\hat{\mathbf{R}}_t^{\leftarrow} + \hat{\mathbf{s}}_{\theta}(\hat{\mathbf{R}}_t^{\leftarrow}, T-t)\right)dt + d\bar{\mathbf{W}}_t, \quad \text{with } \hat{\mathbf{R}}_0^{\leftarrow} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d) \text{ or } P_T, \end{cases} \quad (28)$$

where P_T is the terminal distribution of the forward SDE (1). It holds that

$$\mathbb{E}\|\mathbf{R}_{T-t_0}^{\leftarrow} - \hat{\mathbf{R}}_{T-t_0}^{\leftarrow}\|_2^2 = \mathcal{O}\left((1 + \sigma_{\max}^k) d^{\frac{5}{4}} k^{\frac{k+10}{4}} n^{-\frac{1-\delta(n)}{k+5}} \log^{\frac{5}{2}} n\right). \quad (29)$$

The proof of Lemma 3 is deferred to Appendix C.2.2. By the Cauchy-Schwarz inequality and Lemma 3, we bound $\|\Sigma_{t_0} - \tilde{\Sigma}_{t_0}\|_{\text{op}}$ as well as $\|\tilde{\Sigma}_{t_0} - \check{\Sigma}_{t_0}\|_{\text{op}}$ by $\mathcal{O}(\sqrt{\mathbb{E}\|\mathbf{R}_{T-t_0}^{\leftarrow} - \hat{\mathbf{R}}_{T-t_0}^{\leftarrow}\|_2^2} \cdot (\sqrt{\mathbb{E}\|\mathbf{R}_{T-t_0}^{\leftarrow}\|_2^2} + \sqrt{\mathbb{E}\|\hat{\mathbf{R}}_{T-t_0}^{\leftarrow}\|_2^2}))$, where the second moments of $\mathbf{R}_{T-t_0}^{\leftarrow}$ and $\hat{\mathbf{R}}_{T-t_0}^{\leftarrow}$ are clearly finite. Putting together all the error terms, we complete the proof of Lemma 2.

Highlights of Technical Novelties. We further compare our Lemmas 2 and 3 to the closest related work of Chen et al. (2023). In particular, our Lemma 2 gives an error bound for the

subspace estimation using generated samples from Algorithm 1. Due to the presence of noise ε , our theoretical analysis is established in the entire d -dimensional space, while Chen et al. (2023) reduce the analysis to the k -dimensional subspace. Thus, obtaining a weak dependence on d becomes more challenging. We address this by establishing Lemma 3.

Furthermore, Lemma 3 quantifies the distributional discrepancy between the backward processes governed by the true score function $\nabla \log p_t$ and the neural score estimator $\hat{\mathbf{s}}_\theta$ at time $T - t_0$, thereby linking the score estimation error to the distribution mismatch error. Unlike Chen et al. (2023), which projects the time-reverse process onto a time-invariant subspace, our framework introduces a time-varying latent subspace induced by $\Lambda_t^{-1/2}\beta$. This added complexity presents analytical challenges in characterizing the relationship between $\Lambda_t^{-1/2}\beta\mathbf{R}_t^\leftarrow$ and $\Lambda_t^{-1/2}\beta\hat{\mathbf{R}}_t^\leftarrow$. To address this, we employ a coupling argument that enables a more generic analysis of the backward processes $\mathbf{R}_{T-t_0}^\leftarrow$ and $\hat{\mathbf{R}}_{T-t_0}^\leftarrow$ in (3) and (4).

6 Numerical Study with Synthetic Data

In this section, we use our diffusion factor model to learn high-dimensional asset returns under a synthetic factor model setup. We evaluate numerically its effectiveness in terms of recovering both the latent subspace and the return distribution (as in Theorem 3).

To simulate a practically challenging scenario, we set the number of assets to be $d = 2^{11} = 2048$ and the number of latent factors to be $k = 16$. Appendix D provides more details on how we construct simulated returns. We denote μ and Σ as the ground truth mean and covariance matrix of returns. Similarly, we denote μ_{Diff} and Σ_{Diff} as the mean and covariance matrix estimated using diffusion-generated data, and μ_{Emp} and Σ_{Emp} as the empirical mean and covariance matrix estimated using training data.

Latent Subspace Recovery. We compare the following two methods to recover the latent subspace:

1. **Diff Method:** Our proposed diffusion factor model—we first estimate the return distribution using our diffusion factor model trained on the training dataset, then generate a large set of new data, and finally apply principal component analysis (PCA) on the generated data to estimate the eigenvalues and eigenspaces.
2. **Emp Method:** A naïve PCA method—we directly perform PCA on the training data and extract the leading eigenvalues and eigenspaces.

We denote $\{\lambda_i\}_{1 \leq i \leq k}$ as the top- k eigenvalues and $\mathbf{U}\mathbf{U}^\top$ as the leading k -dimensional principal components of the ground-truth Σ . We perform SVD on Σ_{Diff} (resp. Σ_{Emp}) to extract the top- k eigenvalues $\{\lambda_i^{\text{Diff}}\}_{1 \leq i \leq k}$ (resp. $\{\lambda_i^{\text{Emp}}\}_{1 \leq i \leq k}$) and the leading k -dimensional principal components $(\mathbf{U}\mathbf{U}^\top)_{\text{Diff}}$ (resp. $(\mathbf{U}\mathbf{U}^\top)_{\text{Emp}}$).

To assess the accuracy of the eigenvalue estimation, we compute the ℓ^1 relative error for Diff Method and Emp Method as

$$\text{Diff RE}_1 = \frac{1}{k} \sum_{i=1}^k \left| \frac{\lambda_i^{\text{Diff}}}{\lambda_i} - 1 \right| \quad \text{and} \quad \text{Emp RE}_1 = \frac{1}{k} \sum_{i=1}^k \left| \frac{\lambda_i^{\text{Emp}}}{\lambda_i} - 1 \right|. \quad (30)$$